P-C-438a
April 26, 1968
SUPERSEDING
Fed. Spec. P-C-438
March 7, 1963

FEDERAL SPECIFICATION

CLEANING COMPOUND, OPTICAL LENS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type and grade of water-soluble, synthetic, nonionic detergent for cleaning optical lens and elements in optical shops (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Copies of Military Specifications and Standards required by contractors connection with specific procurement functions should be obtained from procuring activity or as directed by the contracting officer.)

.2 Other publications. The following documents form a part of this cification to the extent specified herein. Unless otherwise indicated, issue in effect on date of invitation for bids or request for proposal ll apply.

American Society for Testing and Materials (ASTM) Standards:

D95-62 - Method of Test for Water in Petroleum and other Bituminous Materials.
D941-55 - Method of Test for Density and Specific Gravity of Liquids by Lipkin Bicapillary Pycnometer.
D1331-56 - Methods of Test for Surface and Interfacils Tension of Solutions of Surface-Active Agents.
D1638-61T - Method of Testing Urethane Foam Raw Materials.

Application for copies should be addressed to the American Society for ting and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards generally available for reference from libraries. They are also tributed among technical groups and using Federal agencies.)

. REQUIREMENTS

.1 Material. The cleaning compound shall consist of a 99-100 percent ive nonionic synthetic detergent.

.1.2 Compatibility. No cloudiness, precipitation, or sediment shall formed when the detergent is mixed with anionic and cationic wetting nts when tested as specified in 4.3.2.

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3.2 Color and odor. The cleaning compound shall be a water-white to light straw colored liquid. It shall not have an objectionable odor.

3.3 Chemical and physical properties. The cleaning compound shall conform to the requirements of table I when tested as specified in 4.3.2.

TABLE I. Chemical and physical properties

Requirements	Minimum	Maximum
Water content, percent by weight	---	1.0
Ash content, percent by weight	---	0.25
Saponification number, mg. KOH/g. sample	---	5.0
Hydroxyl number, mg. KOH/g. sample	---	100
Cloud point °F. (°C.)	120°F. (48.9°C.)	160°F. (71°C.)
pH at 25°C., 1 percent solution	6.0	8.0
Surface tension at 25°C. 0.01 percent solution dynes per cm.	---	35
Specific gravity at 25°C./25°C.	1.00	1.10
Biodegradability, percent	80.0	---

3.4 Labeling. Each unit container of cleaning compound shall be durably and legibly marked with the following information:

CLEANING COMPOUND, OPTICAL LENS

Instructions for Use:

Preparation of Cleaning Solution

A solution suitable for cleaning glass and plastic optical lens and other optical elements not affected by water shall be prepared by dissolving 1 ounce of this cleaning compound in a small amount of warm water and making up to 1 gallon with distilled or clear fresh water. Best results will be obtained by adding the cleaning compound in small amounts to the water with vigorous stirring.

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CLEANING PROCEDURE

1. Rinse the lens in clean running water to wet surface and remove particles that might scratch the polished surface.

2. Apply the prepared solution to the surface to be cleaned using a wad of absorbent cotton or a soft clean cloth, free from sizing.

3. Rinse the surfaces in distilled water.

4. Repeat steps 2 and 3 to remove stubborn soils.

5 Workmanship. The cleaning compound shall be a clear, homogeneous uid, free from foreign matter.

QUALITY ASSURANCE PROVISIONS

1 Responsibility for inspection. Unless otherwise specified in the ract or purchase order, the supplier is responsible for the performance ll inspection requirements as specified herein. Except as otherwise ified, the supplier may utilize his own facilities or any commercial ratory acceptable to the Government. The Government reserves the t to perform any of the inspections set forth in the specification e such inspections are deemed necessary to assure that supplies and ices conform to prescribed requirements.

1.1 Certificate of compliance. Where certificates of compliance submitted, the Government reserves the right to check test such items etermine the validity of the certification.

2 Inspection. Sampling for inspection shall be performed in rdance with MIL-STD-105, except when otherwise indicated hereinafter. purposes of sampling, an inspection lot for examination and tests l consist of all the end item prepared in a single batch and itted for inspection at one time.

2.1 Inspection of materials and components. In accordance with 4.1 e, components and materials shall be inspected and tested in accordance all the requirements of referenced specifications, drawings and dards unless otherwise excluded, amended or qualified in this specification pplicable purchase documents.

3 Inspection of the end item.

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4.3.1 Examination of the end item. The end item shall be examined for the defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.1.4. Each lot of end item shall be examined for visual, net contents, and preparation for delivery defects. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled unit containers of the same capacity for the examinations in 4.3.1.1 and 4.3.1.2, and in units of shipping containers for examination under 4.3.1.3.

4.3.1.1 Examination for visual defects. The sample unit for this examination shall be one filled unit container.

Examine	Defect
Construction	Not as specified (see 5.1). Container broken, chipped, dirty, and leaking of contents.
Closure	Not type specified; imperfect.
Seal	Not type specified.
Labeling	Missing, illegible. Does not contain labeling information specified in 3.4.
Contents	Not clear, homogeneous liquid. Contains foreign matter.

4.3.1.2 Examination for net contents. The sample unit for this examination shall be one filled unit container. The lot shall be unacceptable if the average net content per container, for all sample units examined, is less than required.

4.3.1.3 Examination of preparation for delivery. An examination shall be made to determine that material, contents, and marking comply with the requirements of section 5 of this specification. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Closed shipping containers shall be examined for closure defects. The lot size shall be the number of shipping containers in the end item inspection lot.

Examine	Defect
Material	Any nonconforming component; component missing, damaged, or otherwise defective. Inadequate application of components, such as: incomplete closure of container flaps; loose or inadequate sealing, strapping, or stapling.

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ine	Defect
ents	Bulged or distorted container. Number of intermediate packages per shipping container is more or less than required. Number of primary containers per intermediate package is less than specified.
ing (exterior . interior)	Omitted, incorrect, incomplete, illegible, not in accordance with contract requirements.

3.1.4 Inspection levels and acceptable quality levels (AQLs) for inations. The inspection levels, for determining the sample size, the acceptable quality levels (AQLs), expressed in defects per 100 s, shall be as follows:

Examination paragraph	Inspection level	AQLs
4.3.1.1	I	2.5
4.3.1.2	S-2	---
4.3.1.3	S-2	4.0

3.2 Testing of the end item. The end item shall be tested for the acteristics specified in table I in accordance with test methods ified in table II. The lot size shall be expressed in quarts of ning compound. The sample unit for testing shall be 1 quart composite ined by combining equal portions from samples selected at random ughout the lot in the frequency indicated below. The composite shall laced in a clean, dry, sealed quart container. Care shall be exercised revent contamination or alteration of the cleaning compound during the ling, compositing, storage and testing. All test reports shall contain individual values utilized in expressing the final results. The lot l be unacceptable if the composite fails to meet any test requirements ified.

Lot size (quarts)	Sample frequency
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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Characteristic	Specification reference		Requirements applicable to	Number determinations per unit	Results reported as	
	Requirement	Test method	Composite sample		Pass or fail 1/	Numerically to nearest
Water content, percent by weight	Table I	4.4.3	X	2		0.1 percent
Ash content, percent by weight	Table I	4.4.4	X	2		0.01 percent
Saponification number, mg. KOH/gram sample	Table I	4.4.5	X	2		0.1 percent
Hydroxyl number, mg. KOH/gram sample	Table I	4.4.6	X	2		2 number
Cloud point °F. (°C.)	Table I	4.4.7	X	2		1 degree
pH at 25°C.	Table I	4.4.8	X	2		0.1 pH
Surface tension at 25°C., dynes per cm.	Table I	4.4.9	X	2		1 dyne per cm.
Specific gravity at 25°C./25°C.	Table I	4.4.10	X	2		0.001 unit
Compatibility	3.1.2	4.4.1	X	1	X	
Color and odor	3.2	4.4.2	X	1	X	
Biodegradability, percent	Table I	4.4.11				

1/ If failure is indicated, report description or numerical point of failure, as applicable.

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4.4 Test methods.

4.4.1 Compatibility.

4.4.1.1 Preparation of sample. One gram (g.) of the detergent shall dissolved in 99 grams of distilled water.

4.4.1.2 Procedure. Mix a portion of the 1-percent solution of the tergent under test with an equal portion of a 1-percent solution of each the following:

Sodium lauryl sulfate (anion active).

Cetyldimethylbenzyl ammonium chloride (cation active).

cord any cloudiness or precipitation.

4.4.2 Color and odor. Examine the cleaning compound for color and or.

4.4.3 Water content. The water content shall be determined as ecified in ASTM method D95-62.

4.4.4 Ash content. Accurately weigh 1 ± 0.001 gram of the sample d place into a weighed 30-milliliter porcelain crucible. Heat gently volatilize any water. Gradually increase the temperature, being reful to avoid boiling over and spattering. Finally, ignite to constant ight at 600° C. Cool in a desiccator and weigh.

4.4.4.1 Calculation. Calculate percent ash as follows:

$$\text{Percent ash} = \frac{100\ A}{W}$$

Where: A= g of residue.

W= g of sample.

4.4.5 Saponification number. Accurately weight 5 ± 0.001 grams of the mple into a 300-milliliter Erlenmyer flask and add from a pipette .0 milliliters of approximately half-normal alcoholic potassium hydroxide. at on a water steam bath under reflux for 1 hour. Add three drops of percent phenolphthalein in alcohol and titrate with a standardized lf-normal sulfuric acid. Repeat the procedure on a blank.

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4.4.5.1 Calculation. Calculate the saponification number as the number of milligrams of potassium hydroxide required for 1 gram of the sample as follows:

$$\text{Saponification number} = \frac{(A - B) \times N \times 56.1}{W}$$

Where: A = ml. of standard H_2SO_4 required for blank titration.

B = ml. of standard H_2SO_4 required for sample titration

N = normality of sulfuric acid.

W = sample weight in grams.

4.4.6 **Hydroxyl number.** The hydroxyl number shall be determined as specified in ASTM D1638-61T, method B.

4.4.7 Cloud point. Dissolve 0.5 gram of sample in 100 milliliters of distilled water. Transfer 10 milliliters of this solution to a clean, dry, 25-by 200mm. test tube, and insert a thermometer. The test tube shall then be inserted about half-way in an 800-milliliter beaker nearly full of water. Mechanical stirring of the water shall be provided and low heat shall be applied to raise the temperature slowly. The sample shall not be stirred until a preliminary indication of the cloud point is obtained by turbidity developing in the layers next to the wall of the test tube. From the time this is observed until the solution suddenly becomes cloudy, the sample shall be stirred continuously by the thermometer. The cloud point shall be reported in °F. as the temperature at which cloudiness appears throughout.

4.4.8 pH value. Dissolve 1.0 gram of the sample in freshly boiled and cooled distilled water, dilute to 100 milliliters with freshly boiled and cooled distilled water, and mix. Determine the pH value (to the nearest 0.1) of this solution at 25°C. by means of a suitable pH meter employing a glass electrode.

4.4.9 Surface tension. Dissolve 0.1 gram of sample in freshly boiled and cooled distilled water, dilute to 100 milliliters and mix. Determine the surface tension of this solution at 25°C. in accordance with ASTM D1331-56, method A.

4.4.10 Specific gravity. Determine the specific gravity of the cleaning compound at 25°C. in accordance with ASTM D941-55, procedure A.

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4.11 **Biodegradability.** Until an acceptable method of determining biodegradability of a non-ionic detergent is agreed upon by industry is found acceptable to the Government, the supplier shall furnish a ;ificate of compliance certifying that his product meets the generally :pted criteria for degree of degradation in accordance with a current ıercial practice. The certificate shall be accompanied by actual test ı (field or laboratory report) indicating the current commercial :tice used in making the degradation determination.

PREPARATION FOR DELIVERY

1 Packaging. Packaging shall be level A or C as specified (see 6.2).

1.1 Level A.

1.1.1 Unit packaging. Unit packaging shall be one pint or one ·t, as specified (see 6.2). Containers shall conform to group A, s 1, type e, style 1, grade 1 or 2, closure A, seal A or B of C-186.

1.1.2 **Intermediate packaging.** Twelve pint- or twelve quart-containers .l be intermedietely packaged in accordance with the intermediate :aging requirements of PPP-C-186.

1.2 Level C. The cleaning compound shall be packaged to afford .uate protection against deterioration and physical damage during the ment from the supply source to the first receiving activity. The lier may use his standard practice when it meets this requirement.

2 Packing. Packing shall be level A, B or C as specified (see 6.2).

2.1 Level A. Cleaning compound, packaged as specified in 5.1, shall ·acked in accordance with the level A requirements as specified in C-186.

2.2 Level B. Cleaning compound, packaged as specified in 5.1, l be packed in accordance with the level B requirements as specified PP-C-186.

2.3 Level C. Cleaning compound, packaged as specfied in 5.1, l be packed in a manner to insure carrier acceptance and safe delivery estination at the lowest transportation rate for such supplies. ainers shall be in accordance with rules or regulations of carriers icable to the mode of transportation.

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5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The cleaning compound covered by this specification is intended for use in cleaning plastic and glass lenses, coated or uncoated, and other optical elements. Its principal use shall be in optical repair and maintenance shops.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Selection of the applicable level of packaging and packing (see 5.1 and 5.2).
c. Pint or quart size required (see 5.1.1).

6.3 Standard Pack (civil agencies). The Standard Pack for this commodity shall be as specified in 5.1.1 and 5.2.2. The Standard Pack requirements are intended for use in procurements of stores stock replenishments. Procuring officers should use the Standard Pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot, or immediate storage point for temporary storage, subsequent issue, or shipment to eventual user.

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SPECIFICATION

Cleaning Compound, Optical Lens; P-C-438a

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

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